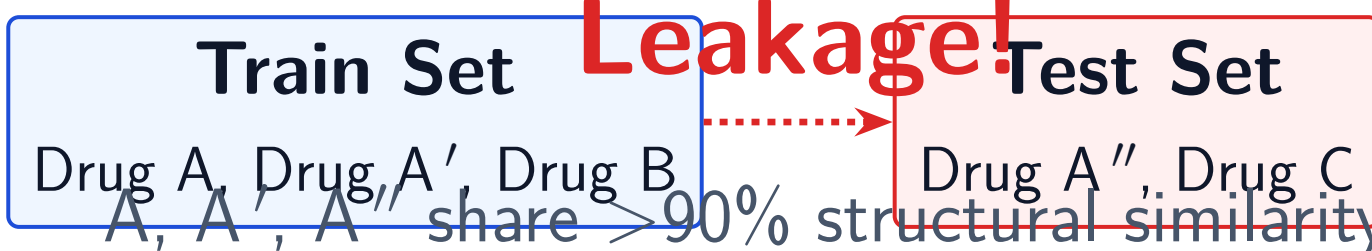


Motivation

Machine learning in drug discovery, materials science, and bioinformatics relies on train/test evaluation. **Random splitting** introduces **data leakage**: structurally similar entities appear in both sets, allowing models to memorize rather than generalize.



Model “predicts” by recalling similar training examples

Random splitting inflates test R^2 by **0.23–0.33** on standard molecular benchmarks (ESOL, Lipophilicity), meaning much of the reported performance comes from leakage rather than genuine generalization.

PALM provides automated, leakage-minimized splitting via both entity-based and cluster-based strategies, accessible through a CLI and web application.

Splitting Strategies: Entity-Based vs Cluster-Based

DataSAIL provides two complementary approaches to prevent leakage:

Strategy	What it prevents
R	Random (baseline)
I1e/I2	Entity-based (1D) Any shared entity on either axis
Scaffold	Scaffold-based Same molecular scaffold across splits
C1e/C2	Cluster-based (1D) Similar entities across splits
C2	Cluster-based (2D) Similar entities on both axes

Entity-based (I*) removes *exact duplicates* — fast, no feature computation needed. Essential when datasets contain repeated entries.

Cluster-based (C*) groups *structurally similar* entities — prevents subtle near-duplicate leakage. Requires features and distance computation.

Scaffold groups molecules by generic Bemis-Murcko ring framework — chemistry-aware, no pairwise distances needed.

Both strategies are complementary: **I*** catches obvious leaks, **C*** catches subtle structural analogues.

Results — 1D Splitting

BBBP dataset (2,050 drug molecules, Morgan fingerprint, Tanimoto distance):

Technique	Mean NN Dist	Zero-Dist Pairs	Dist. Shift
Random (R)	12.3	14	0.042
Identity (I1e)	11.6	8	0.075
Cluster (C1e)	54.8	0	0.112

Identity removes some duplicate pairs (14→8) but does not change mean separation. Cluster achieves **4.5×** greater separation and eliminates **all** identical train–test pairs.

Materials Project (500 crystal structures, MAGPIE descriptors):

Technique	Mean NN Dist	Min NN Dist	Dist. Shift
Random (R)	283.2	67.9	0.071
Cluster (C1e)	494.0	199.5	0.407

Mean separation: **1.7×**. Worst-case (min NN) separation: **2.9×**. Distribution shift is a controlled trade-off — test distributions become broader but never degenerate.

Results — 2D Splitting (Drug–Target Interactions)

Davis DTI (4,000 interactions between 40 drugs and 100 protein targets):

Technique	Drug Overlap	Target Overlap	Coverage
Random (R)	40/40 (100%)	84/84 (100%)	100%
Identity (I2)	0/40 (0%)	0/84 (0%)	45%
Cluster (C1e)	0/40 (0%)	84/84 (100%)	100%
Cluster (C2)	0/40 (0%)	0/84 (0%)	80%

Random splitting leaks **100%** of both drugs and targets. **I2** eliminates all overlap but loses 55% of data. **C1e** eliminates drug overlap while preserving full coverage. **C2** achieves zero overlap on both axes with 80% coverage. The best choice depends on evaluation goals: I2 for strictest deduplication, C1e for drug-centric generalization, C2 for maximum structural separation.

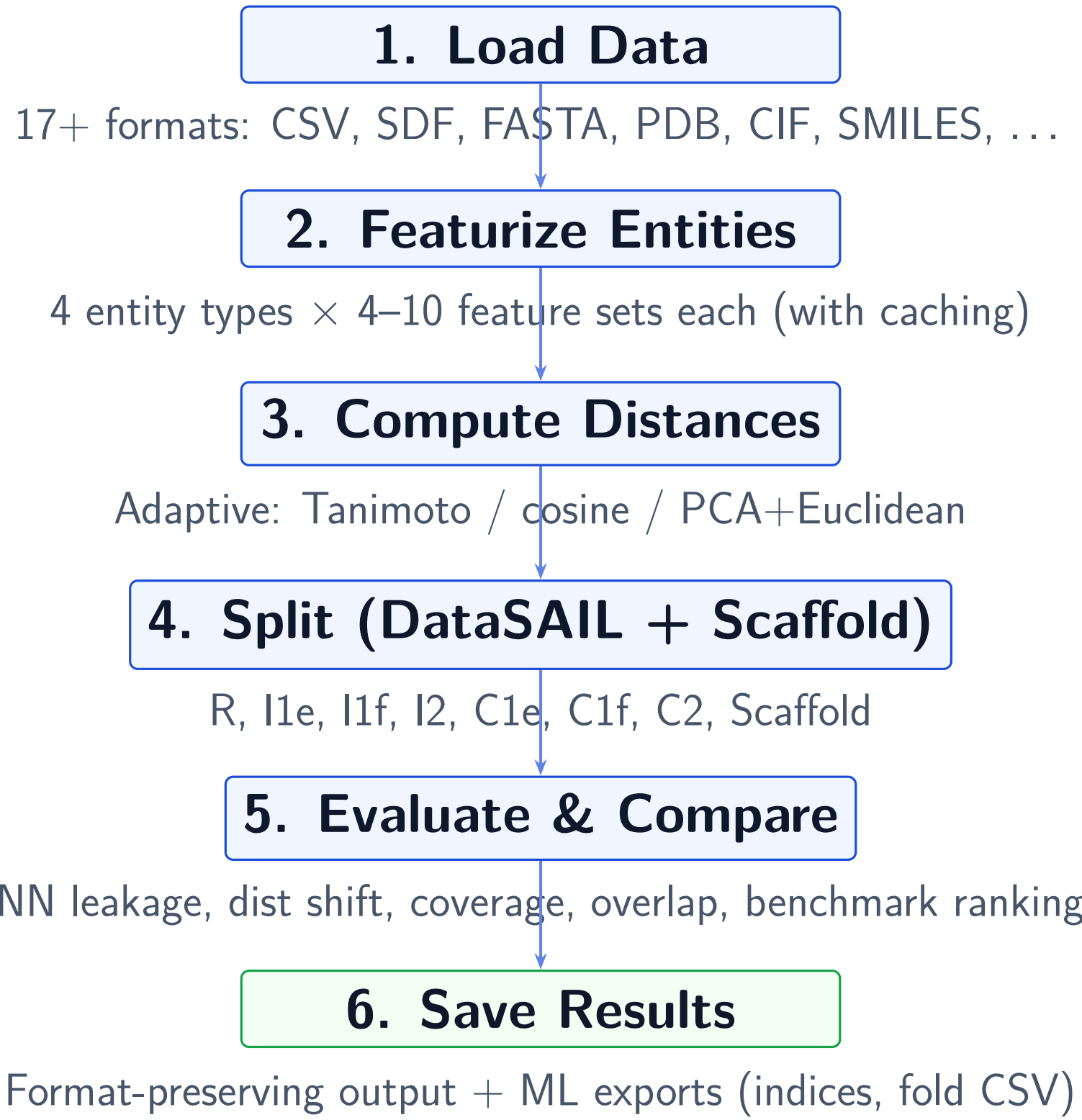
Impact on Model Performance

Random Forest regression (Morgan fingerprint, 500 trees) on molecular property datasets:

Dataset	Split Method	Test R^2	RMSE	MAE
ESOL (1,128 mol)	Random	0.667	1.226	0.906
	Cluster (C1e)	0.333	1.562	1.284
	Inflation	+0.334		
Lipophilicity (1K mol)	Random	0.362	0.928	0.718
	Cluster (C1e)	0.131	1.115	0.868
	Inflation	+0.230		

Random splitting inflates R^2 by **+0.23 to +0.33** because structurally similar molecules leak across sets. Cluster splitting with Tanimoto distance ensures test molecules are structurally novel, producing metrics that reflect real-world generalization to unseen chemical scaffolds.

System Architecture



Entity Types & Feature Sets

Entity Type	Available Feature Sets
Molecule	RDKit descriptors (9 feat), physicochemical (11), composition, Morgan fingerprint (2048-bit ECFP4, uses Tanimoto distance)
Biomolecule	Sequence properties (16 feat), ESM2 protein language model (320–5120 dim), precomputed embeddings
Material	MAGPIE properties (120+), stoichiometry (8), electronic (12), bonding (9), thermodynamic (8), classification (8), mat2vec (200), CrystalINN, SOAP
Gene	Nucleotide composition (11), <i>k</i> -mer frequencies (80), Nucleotide Transformer embeddings (1024–2560 dim), pre-computed

Key Contributions

- Unified framework** spanning four scientific domains with both entity-based and cluster-based splitting
- Scaffold splitting** for molecules using generic Bemis-Murcko ring/linker cores
- Adaptive distance metrics** automatically selected based on feature sparsity
- Format-preserving output** (SDF, FASTA, CIF, etc.) with ML-ready exports
- Automatic benchmark mode** compares all techniques and recommends the best split
- Web application** with auto-detection, guided tooltips, real-time progress, and metrics dashboards

Limitations & Future Directions

- C2 on very sparse, high-dimensional features can reduce coverage — adaptive constraint relaxation partially mitigates this
- Scaffold splitting is molecule-specific; analogues for proteins (fold-based) and materials (crystal system) are future work
- Scaling to >100K entities requires approximate nearest-neighbor distance methods
- Planned: cross-validation folds, temporal splits, batch processing of multiple datasets

Availability

Open source on [PyPI](#) (Python 3.12+) and [conda](#) (Linux, macOS, Windows) installable